

DIFFERENCE BETWEEN BUS-BASED CHIPS AND NoC

Factor	Bus	NoC
Maximum frequency	250MHz	>750MHz
Peak throughput	9GBps	100GBps
Cluster minimum latency	6 cycles @250MHz	6 cycles @ 250MHz
Inter-cluster minimum latency	14-18 cycles @ 250MHz	12 cycles @ 250MHz
System throughput	5GBps	100GBps
Average arbitration latency	42 cycles @ 250MHz	2 cycles @ 250MHz
Gate count	400k	210k
Dynamic power	More	Less
Static power	More	Less

comes with upgraded imaging and camera support, gaming performance boost and advanced connectivity features.

However, the recent entry of the network-on-chip (NoC) is making SoC outdated. The new NoC microcontroller (MCU) can provide more advanced AI integration with low power consumption.

NoC versus SoC

NoC technology applies the theory of computer networking to on-chip communication and brings notable improvements over bus and cross-bar communication architectures. NoC comes in many topologies, many of which are experimental as of 2018. The table above shows a comparison between bus-based chips and NoC.

Replacement of SoC by NoC has resulted in the following advantages:

- Reduced manufacturing cost
- Increased performance
- Reduced time-to-market
- Reduced SoC time-to-volume
- Reduced design risk

War of the NoCs

Quite naturally, demand for NoCs would be increasing because of several advantages, and many companies are now offering these. The next generation of multi-processor SoCs (MPoCs) and chip multi-processors (CMPs) will contain hundreds or even thousands of processors on a single chip. In contrast, NoCs are becoming a promising on-chip communication infrastructure, which is considered an aggressive, long-term

approach for on-chip communication.

While designing NoC, methodologies such as flow control routing, arbitration, quality of service, reliability and task scheduling are considered.

Types of NoC. According to Goosens et al (2003), there are two types of NoCs with respect to quality of service (QoS). These are as follows:

- Best-effort (BE) NoC, which offers no commitment and only ensures completion of communication
- Guaranteed service (GS) NoC, which ensures that some service requirements are accomplished

FlexNoC 4 magic

As nano devices beat the compatibility of mega structures like computers, it is inevitable that some of the trends in the PC processor space will also spill over nano devices. One such trend is NoC. Today, there are several commercial vendors offering NoCs, prominent among them are Arteries, Sonics, Net Speed Systems and Aims Technology.

Arteries is described to be the first commercial company to venture into the NoC market. It invented NoC Interconnect technology and started offering its commercial services in 2006. Today, it has a wide range of customers such as Samsung, Huawei/Hi Silicon, Mobileye, Texas Instruments, Sokle Technology and so on.

“Using Arteries FlexNoC 4 allows us to reduce development time and manage project risk. FlexNoC 4’s advanced QoS and debugging features, combined with its multi-protocol support, allow Samsung SUHD

TVs to reduce power consumption and die area for complex chips with more than 100 IP interfaces,” says Haejoo Jeong, vice president, visual display business, Samsung Electronics.

“Our customers require the latest, most advanced technology, and Arteries FlexNoC 4 Interconnect IP is a necessary addition to our portfolio. Acquiring FlexNoC 4 licence strengthens our technology platform and design services offerings, and provides our customers with more options to future-proof their solutions,” says Stone Peng, president and chief executive officer, Sokle Technology.

With over 20 years of development and more than 250 completed designs, Sonics’ innovative NoC family provides speed, power efficiency and flexibility for chips of all sizes and performance. SonicsHmGN (SGN) is a fourth-generation configurable NoC.

Water Space Mega Chips Corp., Osaka, Japan is not behind in the manufacture of NoCs. On January 29, 2013, it acquired the licence for Arteries FlexNoC Fabric IP to produce next-generation imaging SoCs.

FlexNoC 4: what to focus on

FlexNoC 4 uses SoC interconnect architecture to both accelerate timing closure and improve QoS by using less slack to meet timing, further reducing SoC silicon area and improving performance.

FlexNoC 4 provides advanced technology such as additional configurable hardware IP, visualisation cockpit, timing estimation engine, automatic pipeline placement and configuration.

FlexNoC 4 is a new MCU that supports multiple wireless standards. It is an advanced platform for the Internet of Things (IoT) applications such as Bluetooth Low Energy (BLE), Zigbee, 6LOWPAN and sub-1GHz.

FlexNoC 4 has the following characteristics:

- Layered and scalable architecture
- Flexible and user defined network topology
- Point-to-point connection